Referee response for "Higgs-mode dynamical magnetoelectric effect in a van der Waals multiferroic"

Referee 1

Summary of the Work

The manuscript reports on a study of time-resolved Second Harmonic Generation (SHG) performed on the layered multiferroic compound $CuBr_2$. The authors observe that SHG intensity oscillates following irradiation with femtosecond pulsed laser at a wavelength of 1300 nm or 1650 nm. Notably, the oscillation frequency is shown to be dependent on the polarization configuration:

- Higher-frequency oscillations are observed exclusively in the crossed polarization channels.
- Slower-frequency oscillations appear regardless of the specific polarization settings.

The authors attribute these oscillations to the spin system. Specifically, they argue that the SHG signal monitors ferroelectric polarization induced by the Cu spin cycloid. They rule out phonon-driven oscillations on the basis that known/calculated phonon frequencies are significantly higher than those observed. Ultimately, the authors ascribe the higher-frequency and lower-frequency oscillations to the anisotropy-gapped Goldstone mode and the Higgs mode of the spin system, respectively.

We thank the reviewer for their close review of the manuscript.

Referee comment

Correlation Between SHG Signal and Ferroelectric Polarization:

I am not yet convinced that the SHG signal definitively monitors ferroelectric polarization as proposed. In the present case, where the ferroelectric polarization vector (parallel to the y-axis) lies within the incident plane, the SHG signal should theoretically appear only in the P-out channel, regardless of the incident light polarization. Consequently, the P-in/S-out and S-in/S-out channels should exhibit SHG minima at similar incident plane orientations. However, Figure 4 appears to contradict this:

- A strong SHG intensity is observed at approximately $\phi = \pi/2$ for the P-in/S-out channel.
- Conversely, the intensity peaks at approximately $\phi = 0$ for the S-in/S-out channel.

The authors should clarify this discrepancy to validate their interpretation of the SHG origin.

Our response
Plan:

1. Change figure 1 to show a heatmap plot of the change in SHG intensity vs time
2. Include the equilibrium data as well just to show it but no need for integration regions now Still need to do
3. Update the language to soften any statement that kind of relies on there being no additional contribution Still need to check, maybe there is no such language?
4. Decide what approach to take
   (a) Option 1: Tell the reviewer look, we know there is this additional contribution, this was accounted for in the supplementary material. However, if you look at the change in SHG, it appears to satisfy the constraint they suggested. This is correct, but what's REALLY going on is more complicated; there's heterodyning between the background and the signal
   (b) Option 2: Tell the reviewer the whole story. There is this background. This is what the "pure" signal looks like. This is what the total signal looks like. If do the fitting, you can see that everything works out. However, to show this fitting, we would have to explain that we think it's possible to remove the background signal. That invites questions like, why don't you repeat the measurement on the background free signal?
5. Internally we know that everything works out in detail from Bryan's fits

Option 1 We thank the referee to their close consideration of this point. The referee is correct that the second harmonic generation (SHG) signal, if it were purely due to the ferroelectric polarization vector, should have minima at equivalent positions for the P_{in}–S_{out} and S_{in}–S_{out} polarization channels. The resolution to this comes from something that was briefly touched upon in the Supplementary Material, section S5, where we noted that the static SHG signal we observe is understood to be a combination of the ferroelectric contribution described by equation 1, plus a background contribution that we believe comes from surface adsorbates. Indeed, this additional contribution lifts the nodes in the S_{in}–S_{out} signal and causes the discrepancy noted by the referee.

We note, importantly, that this background contribution does not affect any of the conclusions of this work. Notably, the *change* in the SHG signal after photoexcitation satisfies the referee's constraint; i.e., the P_{in}–P_{out} and P_{in}–S_{out} channels of ΔI_{SHG} are aligned with respect to the incident plane angle ϕ. We have depicted this clearly in the updated manuscript, where Figure 3 (duplicated below) has been modified to show the full angular dependence of the time-resolved SHG (tr-SHG) signal in the four polarization channels. Furthermore, note that the softening of the oscillation frequency close to T_c is direct evidence that the oscillations are related to the phase transition itself and not to the surface adsorbates.

In summary, the referee brings up a very important point and we apologize for the confusion about this in the originally submitted manuscript. We have made the following changes to the manuscript to make this point more clear.
Changes made

1. One potential source of confusion was from the sentence, "The SHG intensity below T_c (Fig. 2) is thus a direct probe of the electric polarization in $CuBr_2$, where the dependence on the polarization of the incident electric field is given by Eq. (1).", on line 98. This sentence is actually trivially incorrect as a background contribution is already visible in the intensity above T_c. We have replaced this sentence by: "The SHG intensity below T_c (Fig. 2) thus probes the electric polarization in $CuBr_2$, where the dependence on the polarization of the incident electric field is given by Eq. (1) heterodyned with a non-ferroelectric contribution from the surface of the sample."

2. A second source of confusion we believe stemmed from not showing the full, angle-resolved time dependence of the SHG intensity in Figure 3. In retrospect we believe this was an error, so we have remade the figure, and have duplicated it here for convenience—see below.

3. We have expanded the discussion of the background signal in the supplementary material, section S5. TODO: This is where we would explain everything about the heterodyning, etc. if Nuh thinks we should do that.

Referee comment

> Interpretation of the Slower Oscillation as a Higgs Mode:
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> I question whether the observed slower oscillation can be strictly attributed to a Higgs mode (i.e., an amplitude fluctuation). If the ordered cycloid component were fluctuating, one would expect the ferroelectric polarization to both rise and fall. However, the data shows that the SHG signal repeatedly decreases and recovers, but does not increase beyond its baseline. The authors should address why the expected "rise" in polarization is absent if this is indeed an amplitude fluctuation mode.

Our response
The rise part of the oscillation being absent is naturally accounted for by a pump-induced heating of the system. This transient increase in temperature will quench the magnitude of ferroelectric polarization. We this expect the Higgs mode oscillation to be about this reduced mean value of polarization. We have updated the main text to explain this, and thank the referee for raising this important point.

Changes made Added the following sentences to the main text: "We note that, in addition to exciting coherent oscillations, the pump pulse is expected to heat the system, resulting in a transient quench of the ferroelectric order

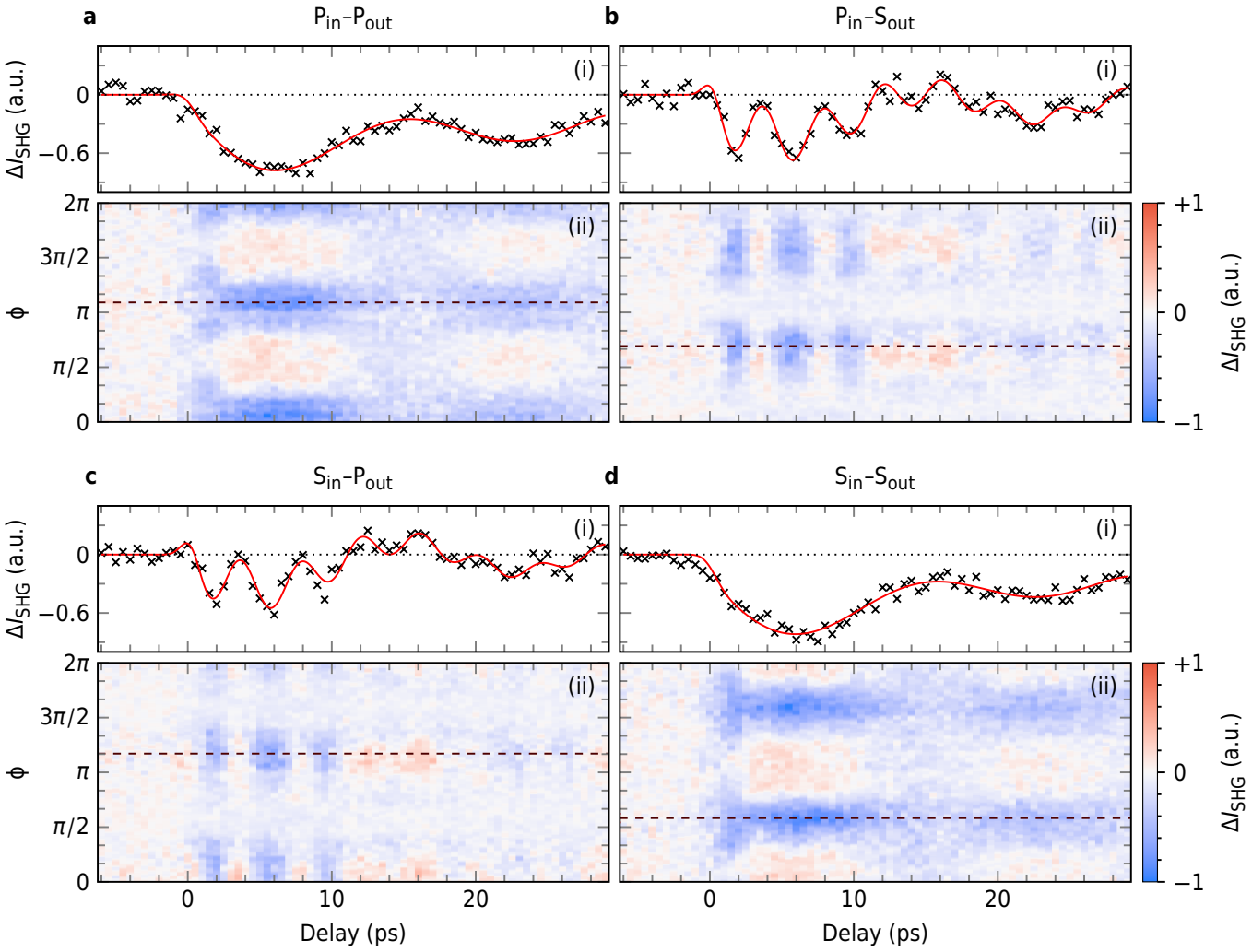


Figure 3: Pump-induced change in the SHG intensity at 15 K in the four polarization channels **a**, P_{in}–P_{out} **b**, P_{in}–S_{out} **c**, S_{in}–P_{out}, and **d**, S_{in}–S_{out}. Upper panels (i) depict line profiles computed by averaging the time-resolved signal in a range ±5° from the dashed lines in the lower panels (ii). Red solid lines depict the best fit to the line profiles using a sum of two harmonic oscillators (Goldstone and Higgs), with a zero amplitude contstraint for the Goldstone mode in the P_{in}–P_{out} and S_{in}–S_{out} channels.

parameter, and hence the SHG intensity. Indeed, the observed oscillations occur about a lower mean value of SHG intensity."

displacive mechanism, cos vs sin? Also, the bandgap is apparently around 2 eV, whereas the pump frequency is 0.7 or so. Nevertheless there can be heating via some high order process, or via a first order process due to mid-gap states. I think we probably don't have to go into details here of how the heating happens.

Referee comment

> Experimental Configuration:
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> The description of the experimental geometry is currently unclear. In the caption for Figure 2, the authors state: "The sample normal is perpendicular to the (001) lattice plane, with the $\phi = 0$ axis coincident with the chain direction in Fig. 1a". Please clarify what is meant by the "$\phi = 0$ axis" and whether it is distinct from the ϕ axis itself.

Our response

We thank the referee for identifying this and apologize for the confusing wording. The proper statement is that the chain axis is contained in the scattering plane when $\phi = 0°$. We have clarified the wording in the main text.

Changes made Replaced the problematic wording in the Fig. 2 caption: "The sample normal is perpendicular to the (001) lattice plane. When $\phi = 0°$, the chain axis of Fig. 1a is contained in the scattering plane."

Math definition verification from Ajesh would be helpful

Referee comment

> Symmetry and Orientation:
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> Since the sample normal is the c*-axis but not the twofold axis (b-axis) of the monoclinic crystal, the sample orientation is not uniquely determined solely by the orientation of the b-axis (chain direction).

Our response

We thank the referee for identifying this ambiguity. Indeed, the referee is correct that the sample orientation is not uniquely determined by the orientation of the b-axis. We have added an additional remark in the methods section to clarify this to the reader. We note that, to our understanding, no statements in the text depend on this ambiguity, nor does it affect any of the conclusions of this work.

Changes made Added the following sentence: "Note that, since the sample normal is the c^* axis but not the twofold axis of the monoclinic crystal, the sample orientation is not uniquely determined by the orientation of the b-axis in the SHG experiment."

I kind of used the refereee's language here, that should be fine right?

Referee comment

> Incident Angle:
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> The incident angle of the fundamental light, θ, must be explicitly stated. Specifically, is θ larger than the angle between the y-axis and the c*-axis? This relationship significantly impacts the resulting SHG signals and their interpretation.

Our response
We thank the referee for noticing this omission. The incident angle ($\theta = 10°$) of the fundamental light is indeed important to state. This value is not larger than the angle between the y-axis and the c*-axis. We have added this value to the methods paragraph.

Changes made Added the following sentence to the methods section: "The angle of incidence of the fundamental light was $\theta = 10°$."

The y-axis and c*-axis are always 90° from each other right? So what is the ref thinking?

Referee comment

> Contextualizing Magnon Excitation:
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> On page 1, the authors state: "Conventionally, the DME is realized by coupling an external electric field to the electric dipole active pseudo-Goldstone magnon". It should be noted that various types of magnons can be excited by oscillating electric fields. For instance, several types of electromagnons were identified in perovskite rare-earth manganese oxides over a decade ago (e.g., N. Kida et al., J. Opt. Soc. Am. B 26, A35 (2009)).

Our response
Plan:

1. This is true.
2. We need to read these papers and understand what is different about the electromagnons the referee is referring to.
3. For example, if they're all transverse electromagnons, we can just change the descriptor in our manuscript to "transverse".

We completely agree that other types of magnons can be excited via electric fields, not just pseudo-Goldstone magnons. We have modifed the phrasing in the revised version of the manuscript.

Changes made Revised the paragraph as follows: "Conventionally, the dynamical magnetoelectric effect (DME) is realized by coupling an external electric

field to an electric-dipole-active magnon, i.e., an electromagnon. A variety of electromagnons have been identified in multiferroic materials, including perovskite rare-earth manganites[1;2], van der Waals (vdW) magnets[3], and others. Two types of electromagnons are relevant to this work. The first is the pseudo-Goldstone magnon (Figs. 1b,c), which becomes electric-dipole active in type-II multiferroics and whose energy is set by the magnetic anisotropy gap. The second is due to the Higgs mode of the ordered multiferroic phase (Fig 1d-1e), which involves fluctuations in the *magnitude* of the spins. This mode is also, in principle, electric-dipole active in type-II multiferroics;[4] however, since it typically resides at a higher energy than the Goldstone modes, it is expected to decay quickly into the latter and remain overdamped.[5]"

Referee comment

> Predicted Signal for Goldstone Mode:
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> If the anisotropy-gapped Goldstone mode of the spin cycloid is excited, the orientation of the ferroelectric polarization should oscillate. In principle, the magnitude of the polarization in its original orientation (along the y-axis) should decrease at a frequency twice as large as the oscillation, while an oscillating electric polarization should appear along the z-axis. It is unclear why such a signal is not observed in the S-in/S-out or P-in/P-out channels.

Our response
Plan:

1. This is easy to refute, we just need Ajesh to write some pretty equations and show that this contribution goes like $(\delta|P|)^2$.

We thank the referee for considering this, this is a point which was not stated clearly in the manuscript. The crucial point is that the amplitudes of the collective oscillations are small compared to the equilibrium values. Working to linear order, we show in the Supplemental Material that, indeed, the anisotropy-gapped pseudo-Goldstone mode does not couple to a longitudinal polarization fluctuation δP_y. This has been noted in Ref. as well. What happens at quadratic order? Let us assume that the polarization follows the spin-ordering via: $\vec{P}_i \propto \hat{x} \times (\vec{S}_i \times \vec{S}_{i+1})$, which holds if the optical phonon frequency is much larger than the electromagnon frequencies (which is true in the present case). When the pseudo-Goldstone magnon is excited, $\vec{S}_i = S_0 \left(\cos\varphi_i \hat{x} + \sin\varphi_i \hat{y} + \theta \sin\varphi_i \hat{z}\right) / \sqrt{1+\theta^2 \sin^2\varphi_i}$, where $\varphi_i \equiv \vec{Q} \cdot \vec{R}_i$. Averaging over all sites, we obtain the induced polarizations: $\delta P_z = \theta P_0$ and $\delta P_y \propto -\frac{1}{2}\theta^2 P_0$. For $\theta(t) = \Theta \cos(\omega_{\mathrm{EM}} t)$, the longitudinal component therefore oscillates at $2\omega_{\mathrm{EM}}$, in agreement with the referee's statement. However, this occurs at quadratic order in the small fluctuation θ, which we expect makes the corresponding signal small in the experiment.

Importantly, we note that this does not affect our arguments for the identification of the Higgs mode, since the Higgs-mode frequency is much smaller than twice the pseudo-Goldstone mode frequency.

Changes made We have emphasized this point in the main text and added details to the Supplementary Material of the revised manuscript.

what is that citation? revisit: any quantitative way?

Referee comment

Supporting Information:

The measurement temperatures must be explicitly labeled in Figure S1.

Our response

Changes made Explicitly labeled the temperatures in Figure S1.

Referee 2

The work presented by Fichera et al picks up two contemporary aspects in research on quantum materials. First, after the magnetoelectric correlations of multiferroics have been investigated for quite some time now, dynamic magnetoelectric correlations move into focus. Second, the magnetic and electric properties may become tightly entangled through fluctuations and criticality in the quantum world with a potential for novel quantum effects, especially in the low-THz frequency regime.

On this background, the authors of the present work report the observation of magnetoelectrically active Higgs modes in a layered multiferroic. Normally, these would decay immediately, but in the special combination of quantum criticality, mode energies, and spin-charge interaction they may become long-lived and thus observable.

While the observation and discussion of the author's claims in the manuscript are quite qualitative, raising the question if explanations alternative to Higgs-mode observation may be possible, the Supplementary Information presents supporting quantitative analysis. In total, I can imagine publication of this work in Nature Materials if my following concerns can be ruled out.

Referee comment

(1) I do not understand the statement on page 2 according to which a purely magnetic c-type electric-dipole term is not permitted by the spiral order, implying it is caused by the induced electric polarization. Since the polarization is

improper, induced by the magnetic order (Eq. 2), the symmetry of the magnetic system should be equal or lower than that of the polar order. This, however, would contradict the quoted statement.

Our response
We thank the referee and apologize for the confusingly worded sentence. The intended point was that the state preserves time-reversal symmetry (combined with translations, as usual in antiferromagnetically ordered systems) and that the SHG tensor does not admit contributions (c-type) that break time-reversal symmetry. It is indeed not the case that the magnetic and polar subsystem possess distinct symmetries. We have clarified this in the main text.

Changes made Replaced the problematic sentence with the clearer statement: "Since the magnetic propagation vector has $k_z = 1/2$, time reversal combined with a lattice translation along $\vec{z}$ is a symmetry of the magnetic state, so time-odd (c-type) contributions to the SHG susceptibility vanish."

Referee comment

(2) Another statement on page 2 is that the Higgs mode should modulate all the SHG susceptibility components equally. This would be true if they are all magnetically induced. However, we clearly have some contributions not coupling to the magnetic order, as the nonzero signal in Fig. 2c above the critical temperature shows. Furthermore, the polarization rotation mentioned at the top of page 6 would mix susceptibilities and modulate them to a different degree even if there were no "non-magnetic" SHG background.

Our response
This is a subtle point and we thank the referee for considering this aspect of the paper so carefully. The referee is correct; in the presence of additional SHG contributions, the total susceptibility,

$$\chi_{ijk}^{(\mathrm{tot})} = \chi_{ijk} + \chi_{ijk}^{(\mathrm{background})}, \tag{1}$$

with χ_{ijk} defined as in the main text (Eq. 1), will be modulated to a different degree by the Higgs mode. However, the assignment of the 0.05 THz mode to the Higgs mode of $CuBr_2$ only depends on the weaker empirical statement given in the subsequent sentence: "The Higgs mode... should thus appear in all four SHG polarization channels (P_{in}–P_{out}, P_{in}–S_{out}, S_{in}–P_{out}, and S_{in}–S_{out}, each of which measures a different linear combination of χ_{ijk} elements)." To see this, first note that, as discussed in the main text, only two excitations are expected in the multiferroic phase of $CuBr_2$—the Higgs mode and the Goldstone mode. Now consider the reverse assignment, where the Higgs mode occurs at 0.2 THz and strongly modulates the P_{in}–S_{out} and S_{in}–P_{out} channels only, leaving the parallel channels untouched. In that case, an exact cancellation between χ_{ijk} and $\chi_{ijk}^{(\mathrm{background})}$ must be happening to quench the oscillation in the P_{in}–P_{out} and S_{in}–S_{out} channels, which is unphysical, especially in light of the fact that the

0.2 THz oscillation was never seen in the parallel channels across many different samples. Thus, the appearance of the 0.05 THz oscillation in all polarization channels at similar magnitude is highly suggestive that this oscillation is due to the amplitude mode of the multiferroic order. Certainly, this is the most definitive evidence that can be provided by tr-SHG, in the presence of a nonzero $\chi_{ijk}^{(\text{background})}$.

We would also like to emphasize again that the argument for the 0.05 THz mode being a Higgs oscillation is supported not only by the polarization dependence of the tr-SHG response, but also by the softening of its frequency with temperature, the small energy scale which precludes any structural component of the modulation, and the exclusion of other potential sources described at length in the main text and supplementary material. Nevertheless, the referee raises an important point and calls attention to an aspect of our explanation that was misleading. To correct this, we have replaced the sentence on page 2 with the more clear statement described above. We hope that this resolves any confusion highlighted by this comment and thank the referee again for their sharp, careful consideration of the reasoning in our paper.

Changes made Replace the affected sentence with: "According to Eq. 1, the Higgs mode should therefore appear in all four SHG polarization channels (P_{in}–P_{out}, P_{in}–S_{out}, S_{in}–P_{out}, and S_{in}–S_{out}, each of which measures a different linear combination of χ_{ijk} elements[6]), even in the presence of a nonzero background signal."

Want to hear what people think of this response??

Referee comment

> (3) I do not see where the van-der-Waals nature of the material studied here matters for the Higgs-mode observation. Also, these systems are known for their novel ferroelectric sliding modes that may occur in the frequency range investigated here. What is their role in the game?

Our response
Plan:

1. All of our theory indeed suggests that interplane coupling plays little role here. So, you could say that the van der Waals-nature of the material studied here does not matter for the Higgs-mode observation.

2. However, the point is that van der Waals materials are both technologically quite interesting for feasibility reasons, and they are just a new family of materials from which new phenomena are frequently being discovered. Therefore, it feels relevant to mention that the discovery occurred in a van der Waals material, even though it is not relevant to the mechanism of the phenomenon we discovered.

3. We can also potentially cook up some reason van der Waals is important? For example, the 1-dimensional chain motif is certainly important here. Does the same motif occur in non-van der Waals materials? Ajesh said he would think about it

4. For the question about ferroelectric sliding modes, Ajesh said he is thinking about it!

In our work, the physical mechanism for ferroelectricity is magneto-electric coupling to a spin-spiral magnetic order, and as such, the vdW nature of the material does not play a direct role in the observations reported here. Separately, however, vdW materials are an unmistakably important class of systems in materials science, especially for the ability to create heterostructures of vdW components in novel device architectures. For this reason, we find it motivating to highlight that the Higgs mode-DME (H-DME) described here appears in a vdW material. However, we agree that, as written, the reason for this inclusion is not clear. To remedy this, we have included a note in the main text to address this source of confusion. We thank the referee for correctly identifying this problem with our paper.

Changes made Added the following sentence: "While not directly relevant to the observed H-DME, the vdW nature of $CuBr_2$ is interesting from an applied perspective."

The referee also correctly points out that there are other mechanisms for how ferroelectricity could arise in vdW materials. In sliding ferroelectrics, such as parallel-stacked bilayers of hexagonal boron nitride or rhombohedral transition-metal dichalcogenides, the monolayers themselves lack inversion symmetry. An inter-layer charge transfer can this occur resulting in an out-of-plane electric polarization. As layers slide relative to each other, the polarization gets modified, and the associated sliding modes are referred to as ferroelectric sliding modes.

This mechanism, however, is absent in $CuBr_2$. Each $CuBr_2$ layer is itself centrosymmetric, with an inversion center at every Cu site. Any two layers related by a translation, including an arbitrary relative sliding, therefore remain related by inversion through the midpoint between them, so interlayer sliding cannot induce a polarization. Both the mechanism of ferroelectricity and the nature of the collective modes in $CuBr_2$ are therefore distinct from those of sliding ferroelectrics. This is supported by our data: both modes soften as the temperature approaches the magnetic transition at T_N and vanish above it, as expected for collective modes of the magnetic order rather than for lattice vibrations.

Should we add this to the supplement?

Referee comment

> (4) Formal mistakes: Insets and labeling in Fig. 3 are much too small. The "dashed lines" in Fig. 4 are dotted lines in my printout.

Our response
Plan:

1. For figure 3, it will probably change a lot.

2. For figure 4, it's easy to fix so we just fix it!

We thank the referee for these suggestions and have fixed the formal mistakes noted.

Still need to figure out what to do with Fig. 3!!!

Referee 3

> This work provides evidence for a Higgs mode that is stabilized by proximity to a quantum critical point. Dynamical measurements in $CuBr_2$ detect a low-energy mode that is surprisingly long lived. The authors interpret this mode terms of a Higgs mode, which is made long-lived because a proximity to a quantum critical point makes the Higgs mode to lie below the Goldstone mode that is made to have a nonzero energy due to magnetic anisotropy. Theoretical speculations along this line have existed before (eg Ref 23).
>
> Still, the experiment is striking and the mechanism brings out the general conceptual importance, viz new dynamical effect is enabled by quantum criticality. The fact that this general physics is happening in a class of materials that have not in the past been highlighted in the context of quantum criticality further elevates the importance. I therefore feel that the work is potentially suitable for publication in a high profile journal like Nature Materials.

Referee comment

> The interpretation rests on $CuBr_2$ to be born nearly quantum critical. Is there any evidence—thermodynamic or otherwise—for this in the material?

Our response

1. We can just answer honestly that there is numerical evidence for the material being proximately quantum critical but no experimental evidence as it is a pretty new/understudied material.

2. In some sense one could argue that the spectrum of collective excitations is a kind of probe of quantum criticality, so that our own observations provide physical evidence.

3. It is essential to note that no conclusions of our work related to assigining the mode we observe to a Higgs mode relies on the material being quantum critical. The quantum criticality is a reasonable, but untested hypothesis intended to motivate future work on this and related materials. Is this strategically sound. also, probably good to clarify our interpretation does not "rest" on criticality. We also point out quasi-1d physics.

We thank the referee for their close consideration of our paper and very helpful comments to improve our manuscript.

The primary evidence for $CuBr_2$ being quantum critical is numerical. Ref. 7 presents a comprehensive study of the spin-1/2 frustrated XXZ chain, which is a good description of the low-energy magnetic Hamiltonian of $CuBr_2$ due to the quasi-1d arrangement of the $CuBr_4$ spin planes. Figure 1 in that reference describes the numerical phase diagram as a function of the ratio of the magnetic exchange parameters J_1/J_2 and the anisotropy gap Δ. From Ajesh: what papers are these numbers from?, $CuBr_2$ lies near $\Delta = 123\,456\,789$, $J_1/J_2 = 123\,456\,789$ in that parameter space, which is close to the quantum critical point separating the observed vector chiral phase from a paramagnetic Haldane dimer phase.

Importantly, however, we would like to stress that our interpretation does not "rest" on $CuBr_2$ being quantum critical. Certainly the identification of the slow SHG oscillation as a coherent Higgs excitation is independent of the system being quantum critical. The fact that $CuBr_2$ is located close to a quantum critical point was invoked both to provide a plausible explanation for the mechanism stabilizing the mode in this material, as well as to motivate future studies into the many aspects of $CuBr_2$ that cannot be understood entirely in a single paper. Given the strong one-dimensionality of $CuBr_2$, as well as the complex phase diagram of the XXZ spin-1/2 model to which $CuBr_2$ belongs, we believe this invocation is reasonable.

Changes made Added a sentence to the last paragraph of the discussion section: "Further studies regarding the quantum critical nature of $CuBr_2$ and related multiferroic compounds will be of great interest."

Revisit: does Nuh think this is strategically OK? Also ask Ajesh for numbers

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Referee comment

> Second, to say that the authors have really "uncovered" a Higgs-mode dynamical magnetoelectric effect seems to me to be rather strong. It seems to me that providing evidence for the effect, or some such weaker statement, is more suitable. This will not diminish the conceptual importance of the work, but will add motivation for further studies along this general direction.

Our response

We thank the referee for this comment and agree that the suggested wording is more appropriate. We have made the suggested edits and greatly appreciate the suggestion.

Changes made Replaced the word "uncover" in the abstract with the phrase "provide evidence for".

Referee comment

> Third, I would like to see remarks about whether this is the first material in which evidence has been provided for the Higgs mode energy to lie below the Goldstone mode energy, or whether related effect has been observed in other materials classes – such as magnets. This again does not take away the freshness of the observed effect in the current context of multiferroic materials for which the relevance of quantum criticality has been considered to be rare.

Our response

We thank the referee for this important comment. Yes, there are a few such examples. A very similar situation is discussed in Ref. 8, where the authors observe a Higgs magnetic excitation in $C_9H_{18}N_2CuBr_4$. There, the Goldstone modes are also gapped by an anisotropy energy, and the Higgs mode energy is reduced due to proximity to a quantum critical point. The authors of that study argue similarly to us that this stabilizes the Higgs mode by shrinking the phase space satisfies the kinematic condition of the decay process. Note that, in $C_9H_{18}N_2CuBr_4$, the Higgs mode must decay into two Goldstone modes to satisfy a spin conservation rule, so the relevant condition is $\Delta_Higgs < 2\Delta_Goldstone$. That is why, in that work, the measured gap energies are $\Delta_Higgs = 0.48\,\mathrm{meV}$ and $\Delta_Goldstone = 0.34\,\mathrm{meV}$, i.e. $\Delta_Higgs > \Delta_Goldstone$. However, the fundamental idea is the same: the energy of the Higgs mode is reduced close to the quantum critical point (QCP) below a threshold set by the Goldstone mode energy, and this stabilizes the mode.

A second example is in $TlCuCl_3$, see Ref. 9. There, the authors observe a longitudinal magnetic excitation whose energy decreases below the transverse mode energy as the QCP is approached by tuning pressure. Its spectral weight and lifetime also increase close to the QCP.

In summary, the situation described in $CuBr_2$ is not unique, and we greatly appreciate the referee making this point. We have updated the main text to make sure this is clear, and thank the referee for helping us strengthen this aspect of our manuscript.

Ajesh we never considered a spin conservation rule? Also do you agree with the sentence I wrote?

Changes made Added the following sentence to the discussion section: "This mechanism—where the phase space available for decay of the Higgs mode is reduced close to the QCP—is exactly what is thought to stabilize Higgs excitations in non-ferroelectric quantum magnets[8], occurring here in a multiferroic material."

References

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